

LECTURE 2 **CROSS-INDUSTRY** **STANDARD PROCESS FOR** **DATA MINING (CRISP-DM)** **&** **PREDICTIVE ANALYTICS I**

LEK HSIANG HUI

OUTLINE

CRISP-DM

Simple Linear Regression

Multi Linear Regression

Coding Scheme for Categorical Variables

CRISP-DM

CRISP-DM

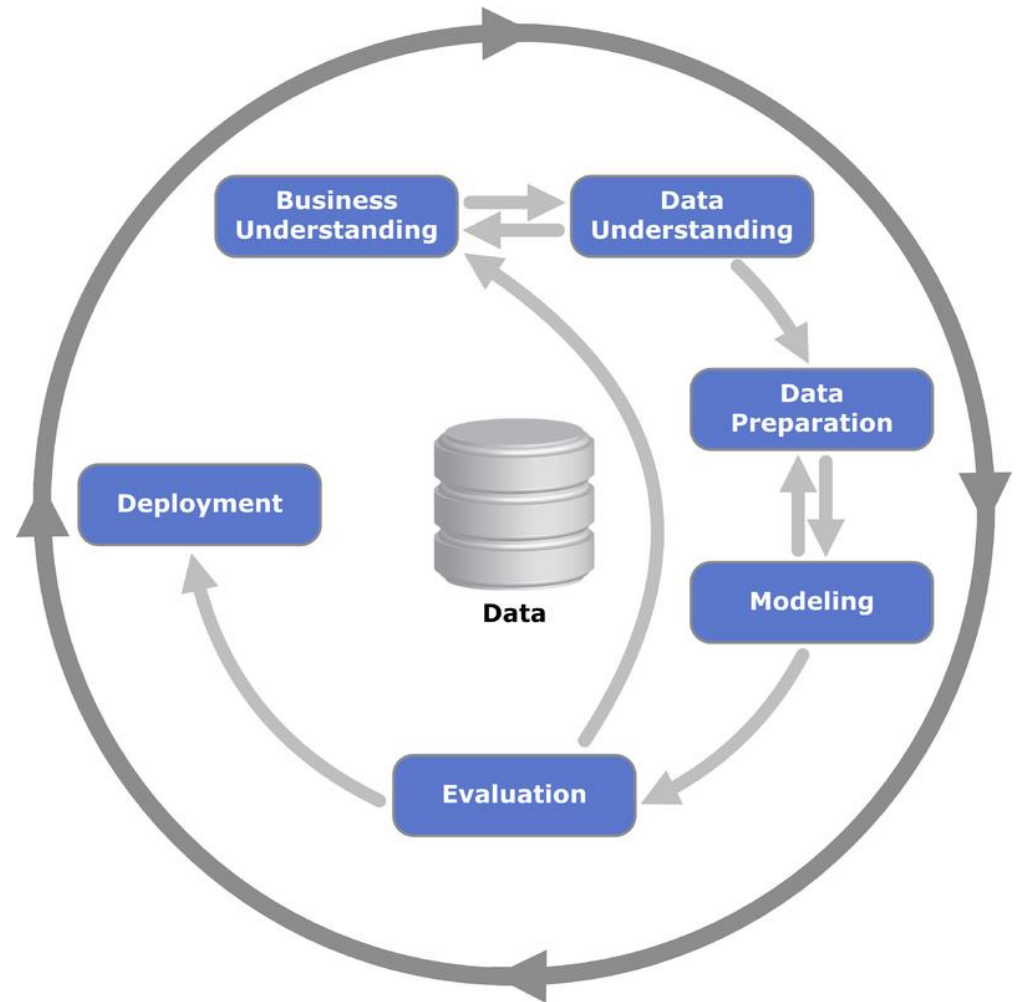
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CRISP-DM

Cross-industry
standard process
for data mining
(**CRISP-DM**) breaks
the process of data mining into 6 major
phases



STEP 1 – BUSINESS UNDERSTANDING

Understand the purpose of the data mining study

- Project objectives
- Requirements of the business
- Rough idea of potential data to use for analysis
- Preliminary plan

Notice that the process starts with the business understanding (i.e. problem)

- It does NOT start with the data!

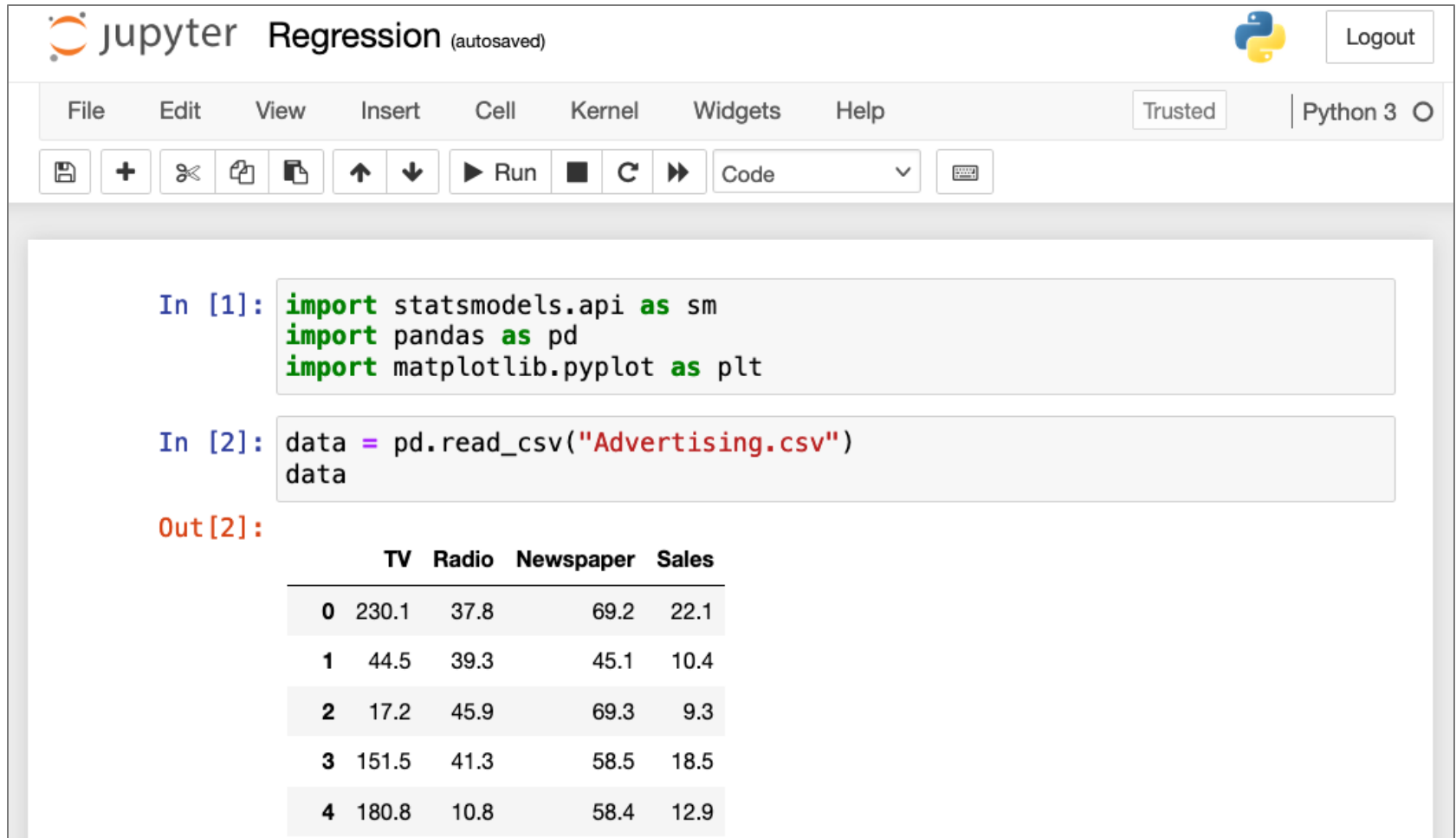
STEP 2 – DATA UNDERSTANDING

Identify the relevant data from the many sources

- Normally: download and use datasets off internet
- Now: learn how to mine the datasets yourself
- Then, perform **Exploratory Data Analysis (EDA)**
 - Perform **statistical analysis**
 - Perform various types of **visualizations**

HANDS-ON: EDA - Q1

Download and access:
[Regression.ipynb](#)



The image shows a Jupyter Notebook interface with the title "Regression (autosaved)". The top bar includes a "Logout" button and a Python 3 environment selector. The menu bar contains "File", "Edit", "View", "Insert", "Cell", "Kernel", "Widgets", and "Help". The toolbar includes icons for saving, adding, deleting, and running cells, along with a "Run" button and a "Code" dropdown menu.

The notebook contains two input cells and one output cell:

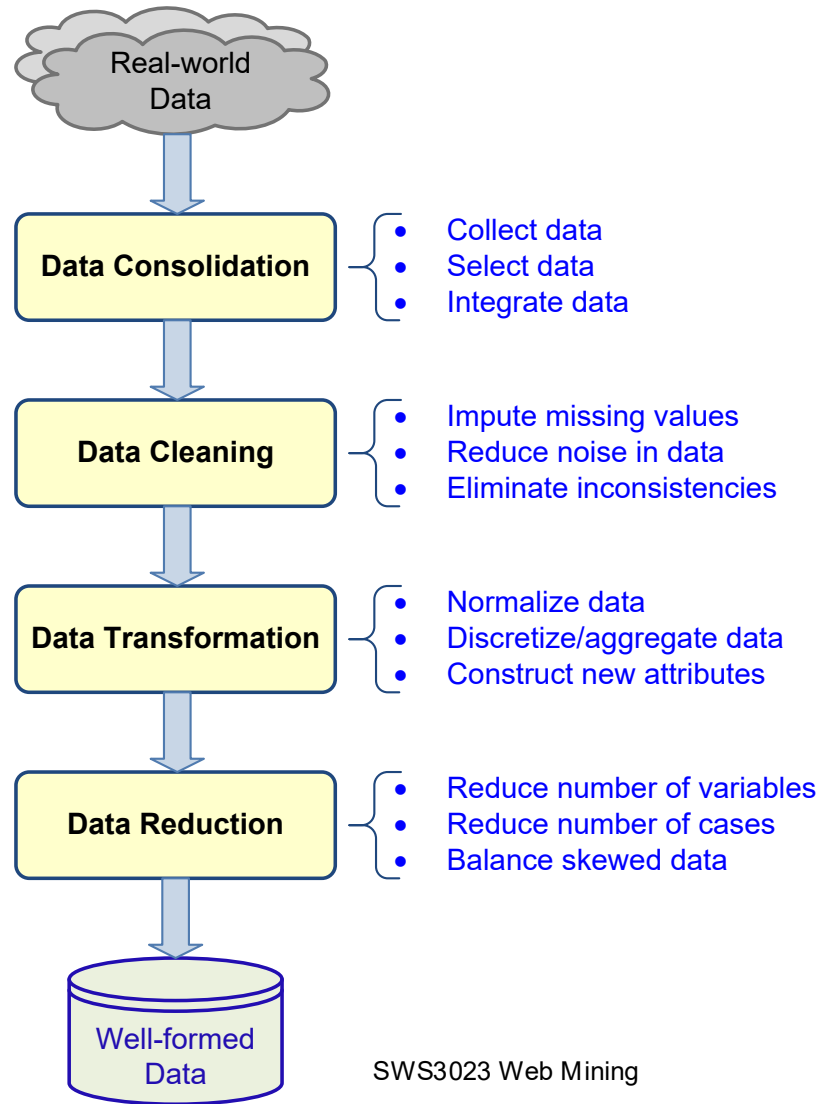
```
In [1]: import statsmodels.api as sm
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: data = pd.read_csv("Advertising.csv")
data
```

The output of the second cell is a table showing the first five rows of the "Advertising.csv" dataset:

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9

STEP 3 – DATA PREPARATION



STEP 4 – MODEL BUILDING

Apply and compare various **data mining techniques**

- Some techniques have specific requirements on the form of data (e.g. need to be numeric)
- Most techniques can only be applied to one type of problem (e.g. classification) while others can be applied for both regression and classification

STEP 5 – TESTING AND EVALUATION

Evaluate the models developed in step 4 (depending on the problem)

- Regression – how far is the prediction from the actual values
- Classification – classification error rates
- Could also have other evaluation methods for other tasks

We usually divide the labeled data into training and testing data and perform K-Fold Cross Validation

STEP 6 – DEPLOYMENT

Development and assessment of model is usually not the end of the project

Depending on the requirements, the deployment phase can be:

- As simple as generating a report
- Or as complex as implementing a system that uses the model for daily operations

Monitoring and maintenance of models

- Over time, the models built may become obsolete

SIMPLE LINEAR REGRESSION

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ADVERTISING EXAMPLE

Suppose we hypothesize that there is a relationship between Sales and amount spend on TV advertisement



SIMPLE LINEAR REGRESSION

Simple linear regression assumes that there is a single predictor variable X and the relationship between the response Y and X is linear

$$Y \approx \beta_0 + \beta_1 X$$

intercept

Slope

This model contains 2 unknown constants that we aim to find

ADVERTISING EXAMPLE

Assume that there is a linear relationship between Sales and amount spend on TV advertisement

$$Sales \approx \beta_0 + \beta_1 TV$$

- Want to see how the spending on TV advertisement can affect Sales
- How to estimate β_0 and β_1 ?
 - Using training data (supervised learning)



TRAINING DATA

Advertising.csv



Thousands \$
spent

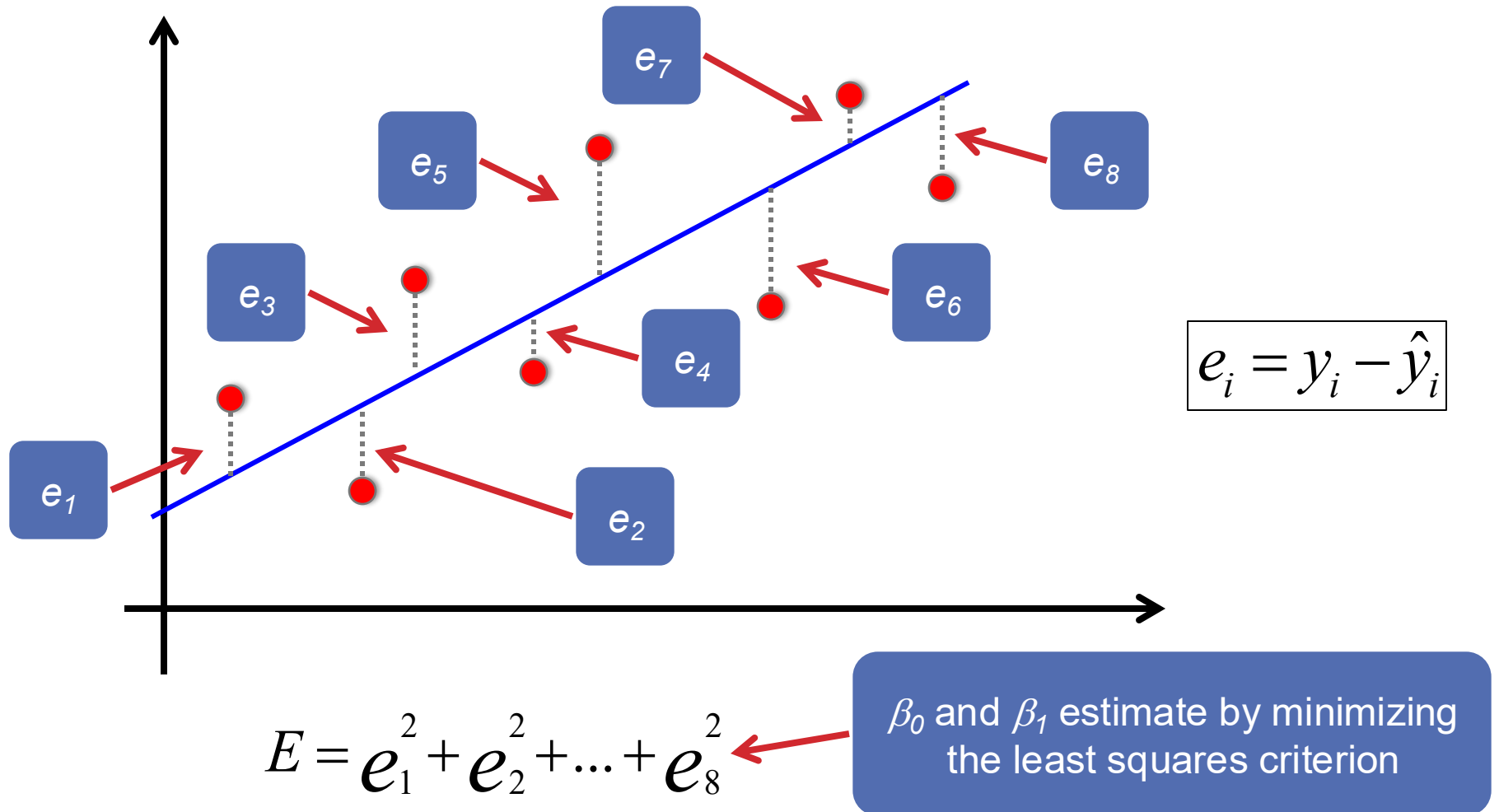
	TV	Sales
1	230.1	22.1
2	44.5	10.4
3	17.2	9.3
4	151.5	18.5
5	180.8	12.9
6	8.7	7.2
7	57.5	11.8
8	120.2	13.2
9	8.6	4.8
10	199.8	10.6
11	66.1	8.6
12	214.7	17.4
13	23.8	9.2
14	97.5	9.7
15	204.1	19
16	195.4	22.4
17	67.8	12.5
18	281.4	24.4
19	69.2	11.3
20	147.3	14.6
21	218.4	18
22	227.4	17.5

200 observations



Thousands
Units sold

LEAST SQUARES CRITERION



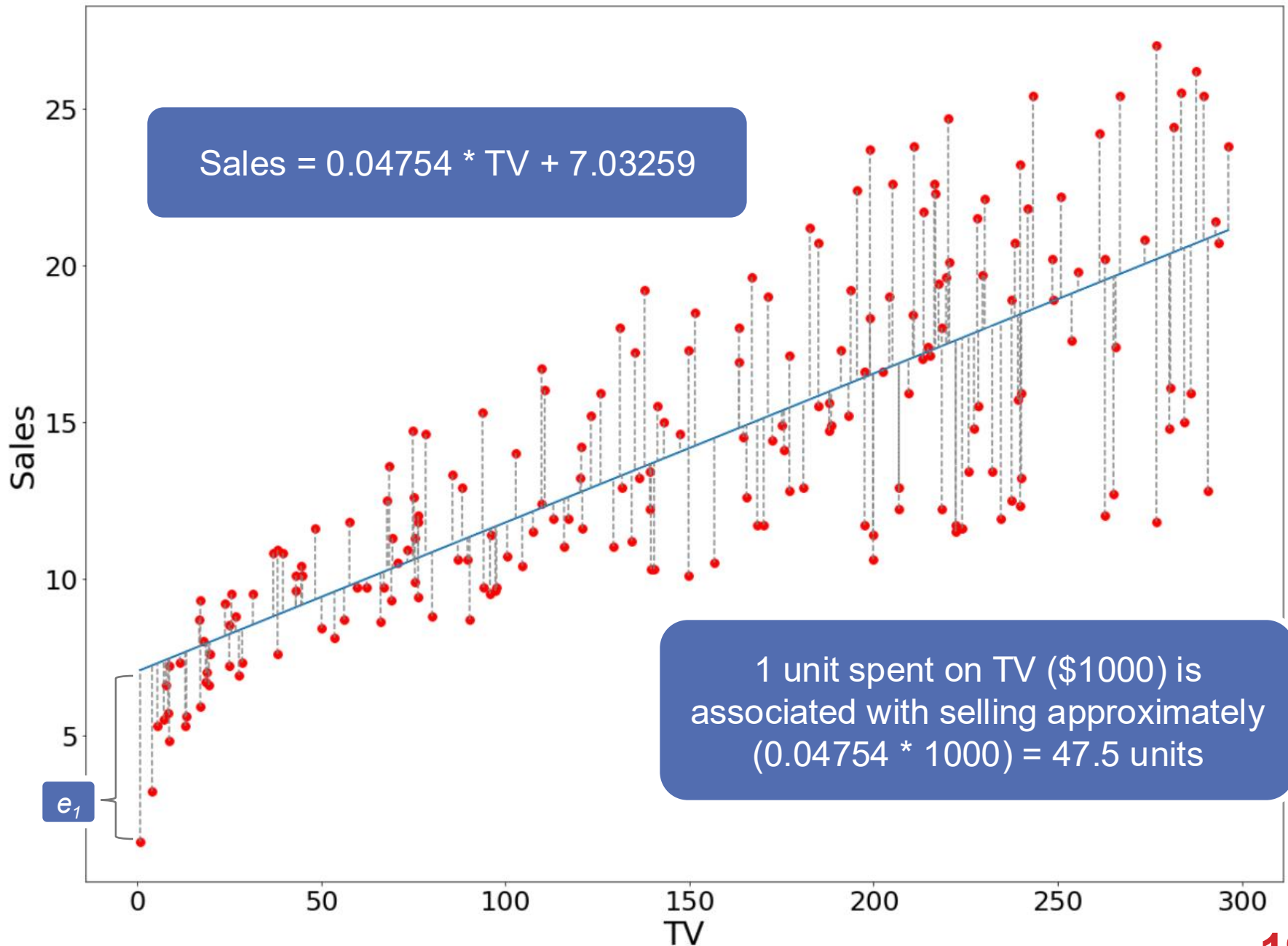
LEAST SQUARES FIT

- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction for Y based on the i^{th} value of X

- **Residual Sum of Squares (RSS)**

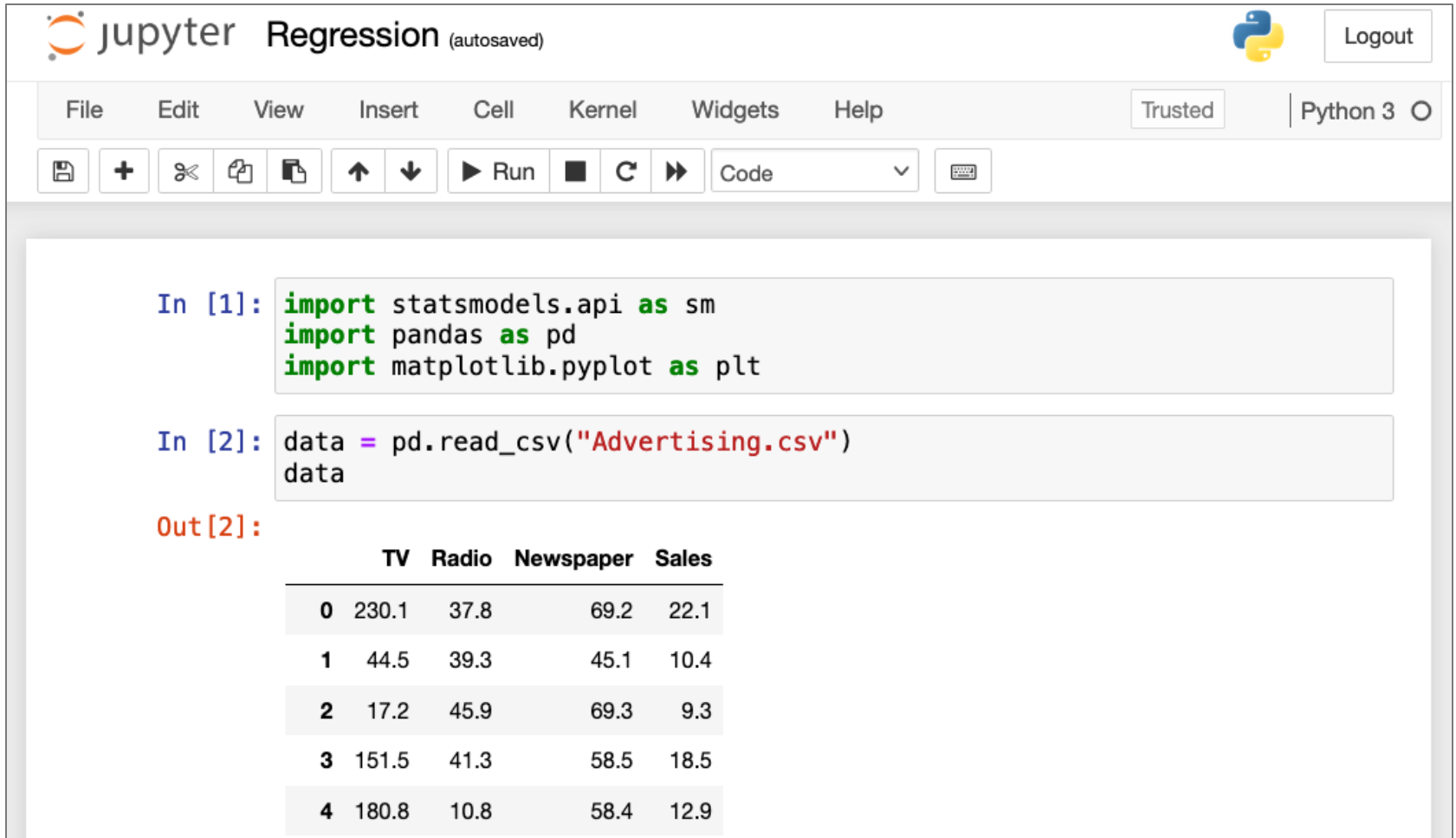
$$RSS = e_1^2 + e_2^2 + \dots + e_n^2$$

- where $e_i = y_i - \hat{y}_i$



HANDS-ON: REGRESSION

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USEFUL PREDICTORS

To determine whether a predictor is useful:

- We check whether the p-value of the coefficient estimate is < 0.05
- Low p-value $\rightarrow$ coefficient estimate is statistically significant

MODEL SUMMARY

OLS Regression Results

Dep. Variable:	Sales	R-squared:	0.612
Model:	OLS	Adj. R-squared:	0.610
Method:	Least Squares	F-statistic:	312.1
Date:	Sat, 12 Jun 2021	Prob (F-statistic):	1.47e-42
Time:	12:49:18	Log-Likelihood:	-519.05
No. Observations:	200	AIC:	1042.
Df Residuals:	198	BIC:	1049.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	7.0326	0.458	15.360	0.000	6.130	7.935
TV	0.0475	0.003	17.668	0.000	0.042	0.053

MEASURE MODEL PERFORMANCE

To measure the quality of fit (of the entire model), we can use:

- R^2
- F-statistics
- Mean Square Error (MSE)

R^2

R^2 measures the proportion of variability in Y that can be explained using X

- Takes value between 0 and 1
- Value close to 0 $\rightarrow$ regression did not explain much of the variability in the response (linear model likely to be wrong)
- In the Advertising dataset, $R^2 \approx 0.61 \rightarrow 0.61$ of the variability in Sales is explained by a linear regression on TV
- What is a good R^2 value depends on the application

ADJUSTED R^2

R^2 will always increase with more variables

- Thus, not really a good way to evaluate the effectiveness of the predictors

Adjusted R^2 factors into the number of predictors in the calculation of R^2 . (Penalize cases where many irrelevant predictors are added)

- Adjusted R^2 is always lesser than R^2
- This is often used instead

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F STATISTICS

F-Statistics is another test to determine whether there is a relationship between the response and the predictors

- Value close to 1 $\rightarrow$ no relationship between the response and predictors
- Value much larger than 1 $\rightarrow$ likely to find relationship between the response and predictors
- More importantly to look at the p-value, whether the F-statistics is significant

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MSE

While R^2 and F-statistics gives a rough idea of how effective is the regression model, it does not tell how much is the error

- The prediction error is sometimes more important

Mean Squared Error (MSE) is able to measure the prediction accuracy/error

$$MSE = \frac{1}{\text{degrees_of_freedom}} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$



Prediction for
observation i based
on our model

MODEL SUMMARY

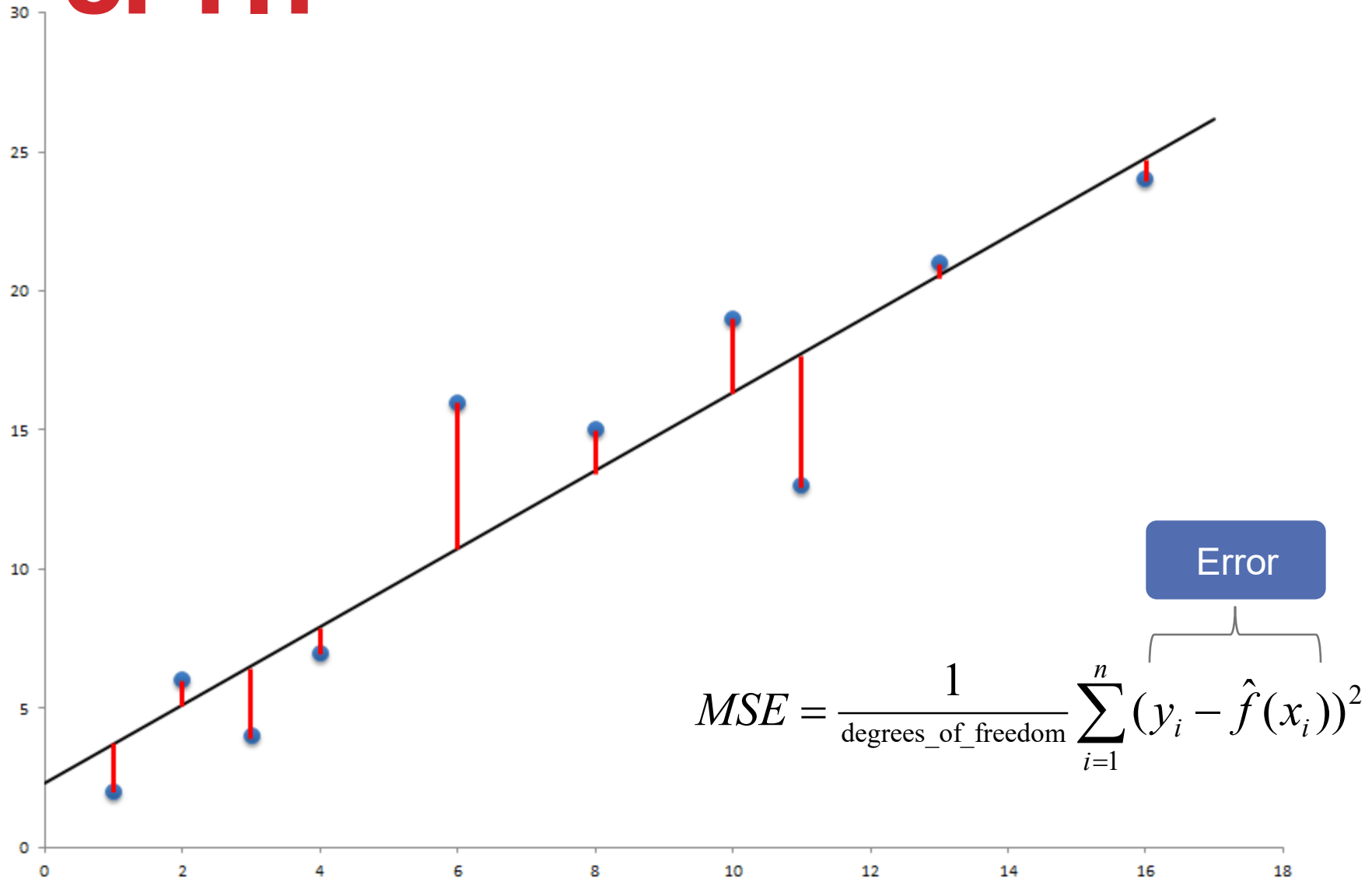
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$(n-2)$ = degrees of freedom
(Lost 2 degrees of freedom because we estimate β_0 and β_1)

MEASURING QUALITY OF FIT



MULTI LINEAR REGRESSION

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MULTI LINEAR REGRESSION



In practice, we would have more than 1 predictor



ADVERTISING EXAMPLE

Suppose we hypothesize that there might be a linear relationship between Sales and amount spend on TV , Radio , Newspaper advertisement

$$Sales \approx \beta_0 + \beta_1 TV + \beta_2 Radio + \beta_3 Newspaper$$

- $\beta_1, \beta_2, \beta_3$ are the coefficients that quantifies the association between TV, Radio, Newspaper spending on the Sales (response)
- β_i is the average effect on Y for one unit increase in X_i while keeping the other predictors fixed

CODING SCHEME FOR CATEGORICAL VARIABLES

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QUALITATIVE PREDICTORS

Regression requires the attributes to be quantitative (i.e. numerical)

Need to specially handle qualitative predictors

	Income	Limit	Rating	Cards	Age	Education	Gender	Student	Married	Ethnicity	Balance
1	14.891	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.025	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.593	7075	514	4	71	11	Male	No	No	Asian	580
4	148.924	9504	681	3	36	11	Female	No	No	Asian	964
5	55.882	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	20.996	3388	259	2	37	12	Female	No	No	African American	203
8	71.408	7114	512	2	87	9	Male	No	No	Asian	872
9	15.125	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.061	6819	491	3	41	19	Female	Yes	Yes	African American	1350

Credit.csv

Qualitative predictors

S3023 Web M

average credit card
debt balance

CODING SCHEME

How to include the gender variable?

2 values: male and female

$$Gender_i = \begin{cases} 1 & \text{if } i\text{th person is female} \\ 0 & \text{if } i\text{th person is male} \end{cases}$$

Supposed we want to include income and gender:

$$Balance_i \approx \beta_0 + \beta_1 Income_i + \beta_2 Gender_i = \begin{cases} \beta_0 + \beta_1 Income_i + \beta_2 & \text{if female} \\ \beta_0 + \beta_1 Income_i & \text{if male} \end{cases}$$

INTERPRETATION

$$Balance_i \approx \beta_0 + \beta_1 Income_i + \beta_2 Gender_i = \begin{cases} \beta_0 + \beta_1 Income_i + \beta_2 & \text{if female} \\ \beta_0 + \beta_1 Income_i & \text{if male} \end{cases}$$

β_2 is the average difference in credit card balance between females and males for a given income level

- Treat males are the “baseline”
- The coding scheme (whether male should be 1 or female should be 1) will not affect the interpretation of the regression

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CODING SCHEME

If there is k ($k \geq 3$) values, create k-1 dummy variables

$$Ethnicity_{ia} = \begin{cases} 1 & \text{if } i\text{th person is Asian} \\ 0 & \text{if } i\text{th person is not Asian} \end{cases}$$

$$Ethnicity_{ic} = \begin{cases} 1 & \text{if } i\text{th person is Caucasian} \\ 0 & \text{if } i\text{th person is not Caucasian} \end{cases}$$

How about African American?

If person is neither Asian nor Caucasian → person is African American

$$Balance_i \approx \beta_0 + \beta_1 Ethnicity_{ia} + \beta_2 Ethnicity_{ic} = \begin{cases} \beta_0 + \beta_1 & \text{if asian} \\ \beta_0 + \beta_2 & \text{if Caucasian} \\ \beta_0 & \text{if African American} \end{cases}$$

WHAT'S NEXT?

Predictive Analytics II